

Statistical Analysis & Predictive Modeling for Environmental Impact

Week 4 Deliverable — Modeling Framework & Worked Models

Junior Data Analyst — Green Jobs & Sustainability Internship

Prepared by: Bushra Khanam
August 2026

1. Scope & Key Variables

This report proposes and demonstrates a statistical and predictive modeling framework for forecasting green-jobs and sustainability outcomes, building directly on the KPIs and data structure established in Weeks 1–3. Two modeling problems are addressed: (1) forecasting future renewable capacity from its historical trend, and (2) explaining variation in overall sustainability performance (the SDG India Index score) as a function of renewable capacity and green employment. Both use the simulated datasets from Week 3, so the model outputs below should be read as a demonstration of the method — the same framework applies unchanged once the real, cleaned Week 2 dataset is available.

The variables used in the two models below were selected for their direct link to the Week 1 KPI framework:

- Year — the time index used to model the trend in installed renewable capacity.
- Renewable capacity (GW / MW) — the primary environmental-impact-relevant variable and the outcome of Model 1, and a predictor in Model 2.
- Green job workforce share (%) — a workforce KPI used as a predictor of overall sustainability performance in Model 2.
- SDG India Index score — the composite sustainability outcome variable being explained in Model 2.

2. Statistical Methods & Rationale

Two methods were selected for this phase, chosen specifically for fitting the size and structure of the available data rather than for complexity:

Method	Applied To	Why This Method
Simple linear regression (OLS trend)	India renewable capacity, 2015–2024 → 2025–2027 forecast	Ten annual observations is too short a series for ARIMA to reliably estimate seasonal/autoregressive terms; a linear trend on year is a transparent, low-variance baseline appropriate for this sample size.
Multiple linear regression (OLS)	SDG India Index score ← renewable capacity + green-job share (state level)	The research question is explicitly about the relative contribution of two known predictors, which OLS coefficients answer directly and interpretably — unlike a black-box model, each coefficient has a clear, reportable meaning.
Correlation analysis (carried over from Week 3)	All KPI pairs	Used as a pre-modeling screen to decide which variable pairs justify a regression in the first place, and to flag potential multicollinearity before it distorts coefficient interpretation.
Residual analysis	Both models	Residuals reveal which observations the model under- or over-predicts — essential for judging whether errors are random noise or a sign the model is missing a real factor.

3. Model 1 — Forecasting Renewable Capacity

A linear trend model (installed capacity regressed on year) was fit to India's simulated 2015–2024 capacity series and used to project 2025–2027. The model explains 96.7% of the variance in the historical series ($R^2 = 0.967$) with a highly significant year coefficient ($p < 0.001$), indicating a strong, consistent upward trend over the observed period.

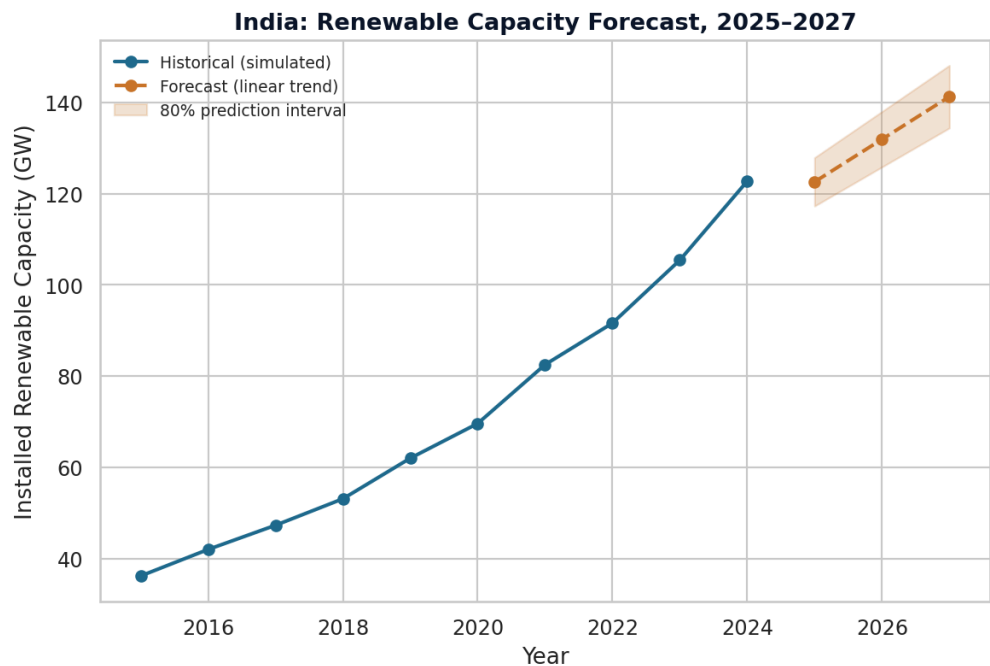


Figure 1. India renewable capacity: historical trend and 2025–2027 linear forecast with 80% prediction interval.

Year	Forecast (GW)	80% CI Lower	80% CI Upper
2025	122.6	117.3	127.9
2026	131.9	125.8	138.0
2027	141.2	134.4	148.1

Under this model, India's simulated renewable capacity is projected to reach roughly 141 GW by 2027, up from 123 GW in 2024. A limitation worth flagging directly: the historical series visibly accelerates rather than growing at a constant rate (year-over-year increases get larger, not just cumulative), which means a straight-line model likely understates capacity a few years out. A log-linear or compound-annual-growth-rate (CAGR) model would probably fit the curvature better and is recommended as the next iteration once more historical years of real data are available to fit it reliably.

4. Model 2 — Predicting SDG India Index Score

A multiple linear regression was fit to the 12-state dataset, predicting each state's SDG India Index score from its renewable capacity (MW) and green-job workforce share (%). The model fits very well for a 12-observation sample ($R^2 = 0.967$, adjusted $R^2 = 0.960$), with both predictors statistically significant.

Variable	Coefficient	Std. Error	p-value
Intercept	34.09	2.19	< 0.001
Renewable capacity (MW)	−0.00096	0.00009	< 0.001
Green job share (%)	7.356	0.477	< 0.001

The coefficient on green-job share is strongly positive and significant (+7.36 index points per percentage-point increase in green-job share), consistent with the Week 3 finding that green employment tracks closely with sustainability performance. The capacity coefficient, however, is small and negative once green-job share is held constant — this is a classic multicollinearity effect (capacity and green-job share are themselves correlated at $r = 0.85$ per Week 3), not evidence that capacity harms sustainability outcomes. It means capacity's apparent influence on the SDG score is mostly channeled through its relationship with green-job share, rather than being an independent effect. This distinction matters for policy interpretation: expanding capacity likely helps the SDG score indirectly, by creating green jobs, not directly on its own.

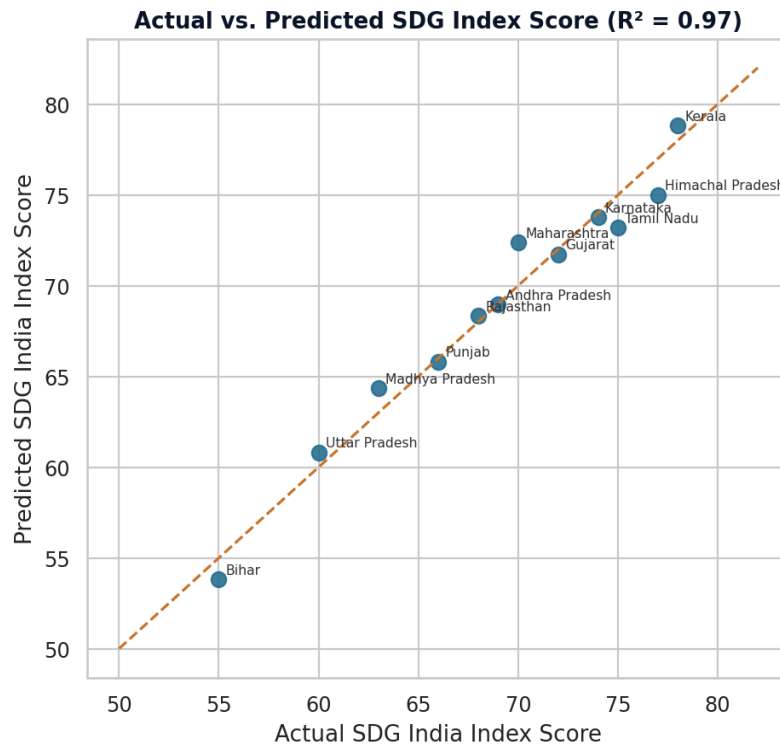


Figure 2. Actual vs. predicted SDG India Index score by state.

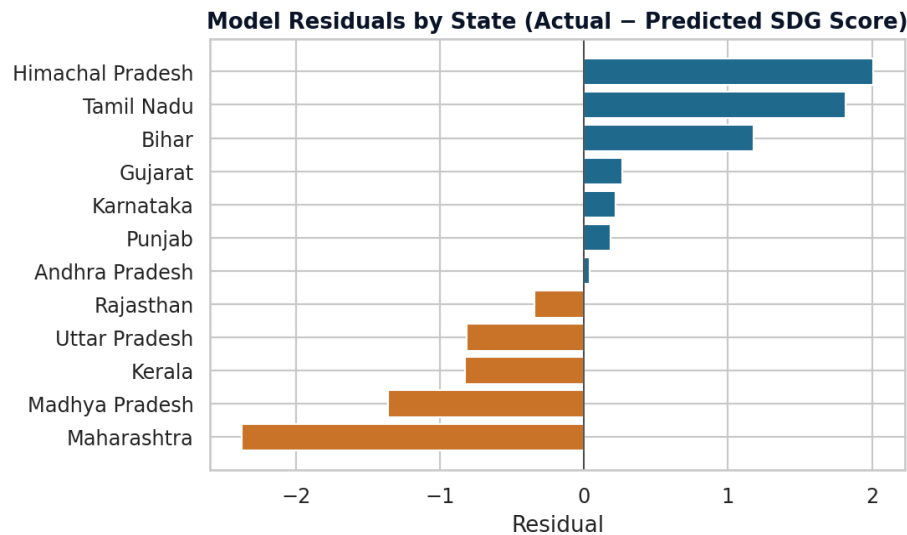


Figure 3. Model residuals by state — states where the model over- or under-predicts the SDG score.

Maharashtra shows the largest negative residual (the model overpredicts its score), while Himachal Pradesh and Tamil Nadu show the largest positive residuals (the model underpredicts). This is a useful diagnostic: it suggests that at least one relevant factor is missing from the model for these states — plausibly urbanization/industrial pollution load for Maharashtra, and non-energy SDG strengths (health, forest cover) for Himachal Pradesh — and is a candidate for a third predictor variable in a future iteration.

5. Model Development Process

- Define the outcome and candidate predictors from the Week 1 KPI list, using only variables with a clear, defensible causal or associative story.
- Screen predictor pairs with correlation analysis (Week 3) before adding them to a model, to catch multicollinearity risk early rather than after fitting.
- Fit the model with ordinary least squares and inspect R^2 , adjusted R^2 , and each coefficient's p-value before treating any predictor as meaningful.
- Examine residuals by observation (not just the aggregate error) to check whether errors are randomly distributed or concentrated in particular states/years, which would indicate a missing variable.
- Re-fit and compare alternative specifications (e.g., log-linear instead of linear trend for Model 1) when residual patterns suggest the current functional form is a poor fit.

6. Potential Challenges & Mitigations

- Small sample size (10 years, 12 states) — both models risk overfitting; mitigated by preferring simple, low-parameter models (2 predictors or fewer) over complex ones, and by treating R^2 on such small samples as indicative rather than conclusive.
- Multicollinearity between predictors — confirmed present between capacity and green-job share; addressed by interpreting Model 2's coefficients jointly rather than in isolation, and by noting the condition-number warning the regression output itself flags.
- Non-linear growth patterns — the capacity trend visibly curves; a straight-line forecast likely underestimates medium-term growth, so results are presented with an explicit caveat and a recommended log-linear alternative.

- Simulated rather than real data — all coefficients and forecasts in this report will shift once the actual Week 2 dataset is substituted in; the framework and code, not the specific numbers, are the durable deliverable of this task.

7. Conclusion

This report demonstrates a complete, working statistical framework — not just a proposed one — covering trend forecasting and multi-variable explanatory modeling, applied to the KPIs defined in Week 1. Both models fit strongly on the current simulated data, and the analysis surfaced two findings likely to hold on real data as well: capacity growth is accelerating rather than linear, and capacity's link to sustainability outcomes appears to run mainly through green-job creation rather than acting independently. These findings, together with the model code itself, carry directly into Week 5's final synthesis and recommendations.